

TOPIC 9: CHANGE OF SUBJECT OF FORMULAE

1. Meaning

Changing the subject of a formula means rearranging it so that a different variable stands alone on one side. The same balancing rules as equations apply: do the same operation to both sides.

2. Involving Addition and Subtraction

From $v = u + at$, make a the subject: $v - u = at \rightarrow a = (v - u)/t$.

3. Involving Multiplication and Division

From $A = lw$, make w the subject: $w = A/l$. From $v = f\lambda$, make λ the subject: $\lambda = v/f$.

4. Involving Roots

From $A = \pi r^2$, make r the subject: $r^2 = A/\pi \rightarrow r = \sqrt{A/\pi}$ (taking the positive root for length).

5. General Strategy

Identify the new subject; use inverse operations to isolate it (undo addition with subtraction, multiplication with division, squares with square roots, etc.); keep both sides balanced; state the final formula clearly.

6. Summary

Changing the subject uses inverse operations to isolate the required variable. Practice with formulae from mathematics and science builds fluency.

End of Topic 9 – Change of Subject of Formulae

— End of JSS 3 Mathematics First Term Notes —